

workspace_1.9.0 - Device Configuration Tool - STM32CubeIDE

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Project Explorer

homework3

- Includes
- Core
- Drivers
- homework3.ioc
- STM32F103C8TX_FLASH

main.c

homework3.ioc

homework3.ioc - Pinout & Configuration

Pinout & Configuration

Clock Configuration

Project Manager

Tools

Software Packs

Pinout

Pinout view

System view

RCC Mode and Configuration

Mode

High Speed Clock (HSE)Crystal/Ceramic Resonator

Low Speed Clock (LSE)Disable

☐ Master Clock Output

Configuration

Reset Configuration

NVIC Settings

Parameter Settings

GPIO Settings

User Constants

Search Signals

Search (Ctrl+F)

Pin Name	Signal on Pin	GPIO output I	GPIO mode
PD0-OSC_IN	RCC_OSC_IN	n/a	n/a
PD1-OSC_OUT	RCC_OSC_OUT	n/a	n/a

STM32F103C8Tx

LQFP48

VDD

VSS

PB9

PB8

BOOT0

PB7

PB6

PB5

PB4

PB3

PA15

PA14

VDD

VSS

PA13

PA12

PA11

PA10

PA9

PA8

PB15

PB14

PB13

PB12

VSS

VDD

PB11

PB10

PB1

PB2

PB0

PA7

PA6

PA5

PA4

PA3

PA2

PA1

PA0

VDDA

VSSA

NRST

PD1

PD0

PC15

PC14

PC13

VBAT

IDE

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homework3.ioc - Clock Configuration

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Resolve Clock Issues

The diagram illustrates the clock configuration for the STM32F103C8TX_FLASH. It shows the flow from input frequencies to various system clocks and peripherals.

Input Frequencies:

- 32.768 KHz (LSE)
- 40 KHz (LSI RC)
- 8 MHz (HSI RC)
- 8 MHz (HSE)

RTC Clock Mux:

- HSE / 128 → HSE_RTC
- LSE → LSE
- LSI → LSI
- Outputs: 40 KHz To RTC (KHz), 40 KHz To IWDG (KHz)

System Clock Mux:

- HSI → HSI
- HSE → HSE
- PLLCLK → PLLCLK
- Enable CSS
- Outputs: 8 MHz To FLITFCLK (MHz), 72 MHz SYSCLK (MHz), 72 MHz HCLK (MHz) (72 MHz max)

PLL Source Mux:

- HSI / 2 → HSI
- HSE / 1 → HSE
- PLL → PLL
- *PLLMul X 9
- USB Prescaler / 1 → 72 MHz To USB (MHz)

APB1 Prescaler:

- HCLK / 1 → 72 MHz HCLK to AHB bus, core, memory and DMA (MHz)
- To Cortex System timer (MHz)
- FCLK (MHz)
- PCLK1 / 2 → 36 MHz max APB1 peripheral clocks (MHz)
- X 2 → 72 MHz APB1 Timer clocks (MHz)

APB2 Prescaler:

- HCLK / 1 → 72 MHz max APB2 peripheral clocks (MHz)
- X 1 → 72 MHz APB2 timer clocks (MHz)
- ADC Prescaler / 2 → 36 MHz To ADC1,2

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Pinout

Search

CategoriesA-Z

System Core

Analog

Timers

- RTC
- TIM1
- TIM2
- TIM3
- TIM4

Connectivity

Computing

Middleware

TIM1 Mode and Configuration

Mode

Slave ModeDisable

Trigger SourceDisable

Clock SourceInternal Clock

Channel1PWM Generation CH1

Channel2Disable

Channel3Disable

Channel4Disable

Combined ChannelsDisable

Activate-Break-Input

Use EXTI as Clearing Source

Configuration

Reset Configuration

NVIC SettingsDMA SettingsGPIO Settings

Parameter SettingsUser Constants

Configure the below parameters :

Search (Ctrl+F)

Counter Settings

- Prescaler (PSC - 16 bits value)71
- Counter ModeUp
- Counter Period (AutoReload Re. 999
- Internal Clock Division (CKD)No Division
- Repetition Counter (RCR - 8 bit...0

Pinout view

System view

RCC_OSC_IN

RCC_OSC_OUT

VBAT

PC13...

PC14...

PC15...

PD0...

PD1...

NRST

VSSA

VDDA

PA0...

PA1

PA2

PA3

PA4

PA5

PA6

PA7

PB0

PB1

PB2

PB10

PB11

VSS

VDD

VDD

VSS

PB9

PB8

BOOT0

PB7

PB6

PB5

PB4

PB3

PA15

PA14

VSS

PA13

PA12

PA11

PA10

PA9

PA8

PB15

PB14

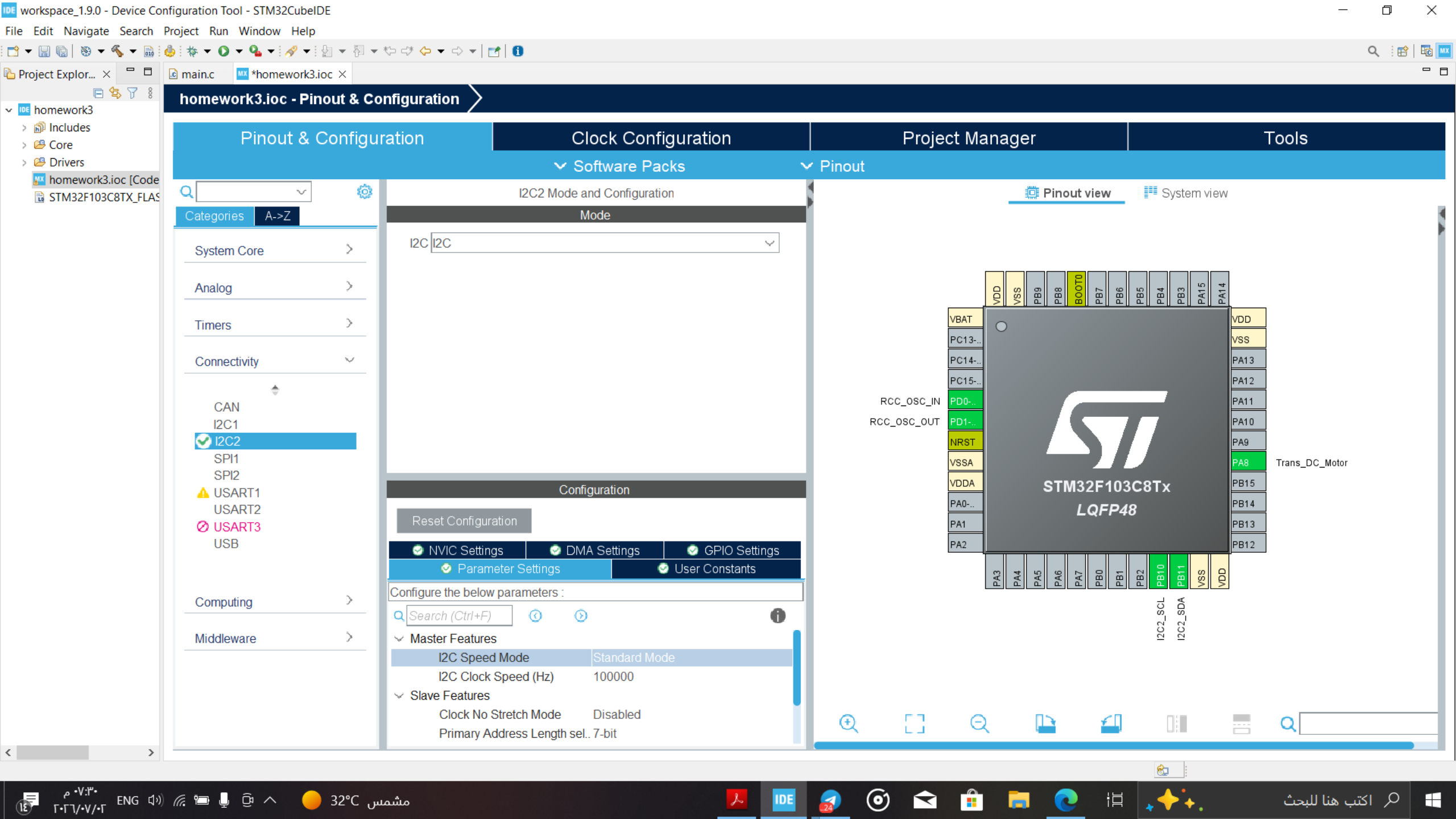
PB13

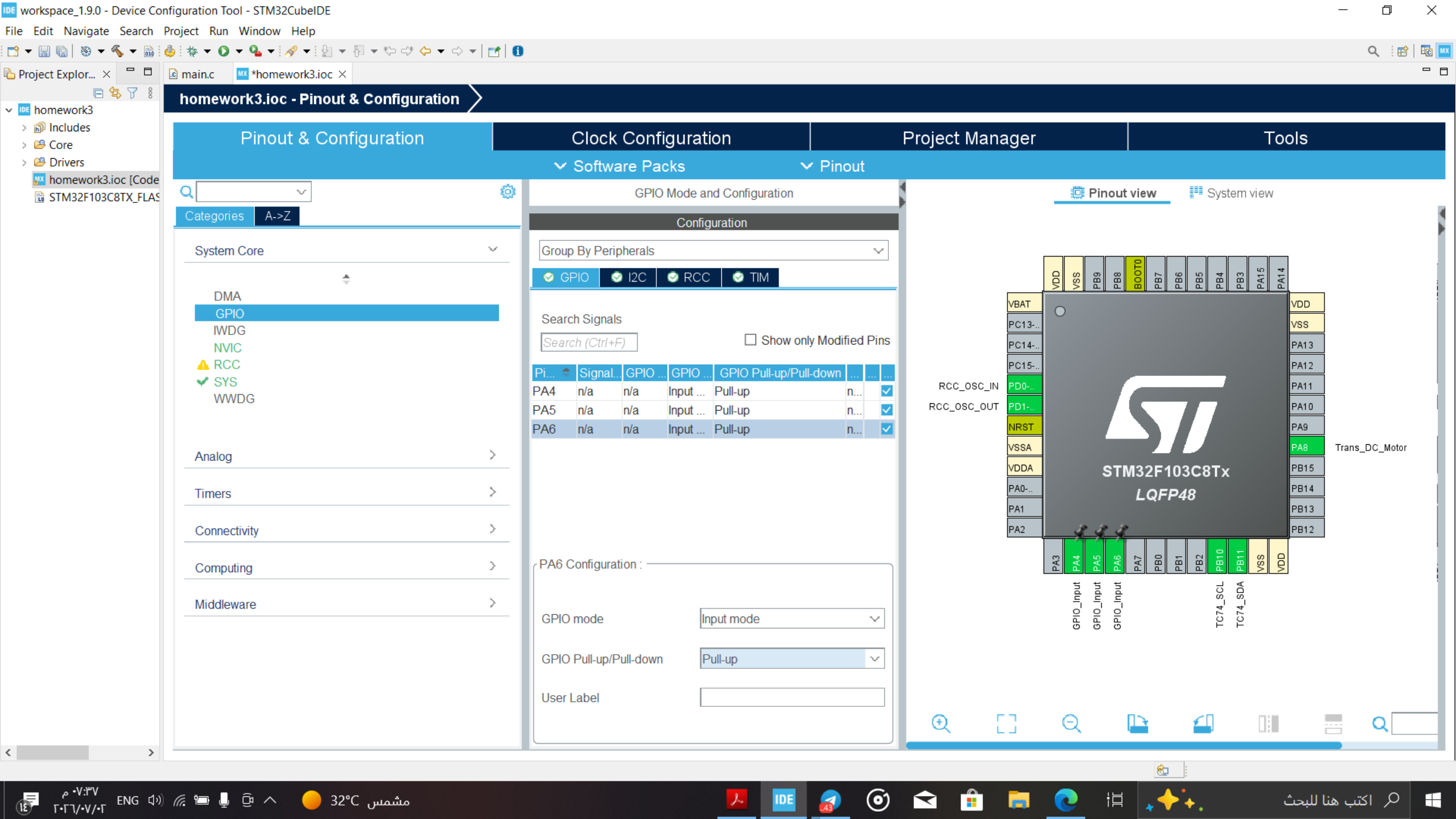
PB12

STM32F103C8Tx

LQFP48

TIM1_CH1





Pinout view

System view

RCC_OSC_IN

RCC_OSC_OUT

Trans_DC_Motor

STM32F103C8Tx

LQFP48

GPIO_Input

GPIO_Input

GPIO_Input

TC74_SCL

TC74_SDA

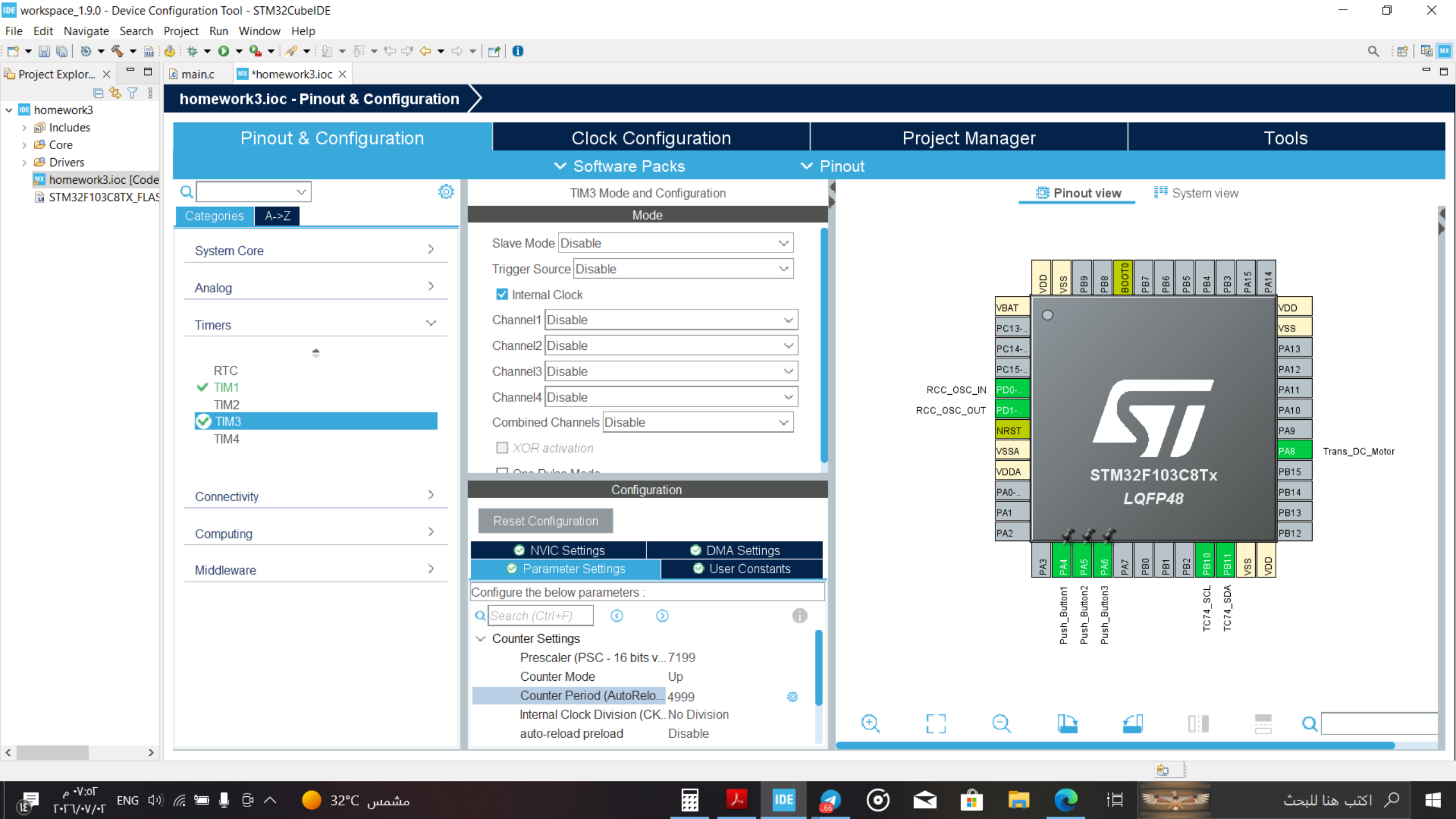
07:37

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NVIC Mode and Configuration

Configuration

NVIC

Code generation

Priority Group 4 bit...

☐ Sort by Preemption Priority and Sub Priority

☐ Sort by...

Search

available interrupts

☒ Force...

NVIC Interrupt Table

Enabled

Preemption

Non maskable interrupt	☒	0
Hard fault interrupt	☒	0
Memory management fault	☒	0
Prefetch fault, memory access fault	☒	0
Undefined instruction or illegal state	☒	0
System service call via SWI instruction	☒	0
Debug monitor	☒	0
Pendable request for system service	☒	0
Time base: System tick timer	☒	15
PVD interrupt through EXTI line 16	☐	0
Flash global interrupt	☐	0
RCC global interrupt	☐	0
TIM1 break interrupt	☐	0
TIM1 update interrupt	☐	0
TIM1 trigger and commutation interrupts	☐	0
TIM1 capture compare interrupt	☐	0
TIM3 global interrupt	☒	0
I2C2 event interrupt	☐	0
I2C2 error interrupt	☐	0

Pinout view

System view

VDD

VSS

PB9

PB8

BOOT0

PB7

PB6

PB5

PB4

PB3

PA15

PA14

VBAT

PC13...

PC14...

PC15...

PD0...

PD1...

NRST

VSSA

VDDA

PA0...

PA1

PA2

PA3

PA4

PA5

PA6

PA7

PB0

PB1

PB2

PB10

PB11

VSS

VDD

PA13

PA12

PA11

PA10

PA9

PA8

PB15

PB14

PB13

PB12

RCC_OSC_IN

RCC_OSC_OUT

Trans_DC_Motor

TC74_SCL

TC74_SDA

Push_Button1

Push_Button2

Push_Button3

STM32F103C8Tx

LQFP48

IDE

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